



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 20/02/2017*

PROBLEM

We observe N samples of the following discrete-time real random process:

$$X[n] = A \cdot s[n] + W[n], \quad n = 0, 1, \dots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a deterministic unknown parameter, $s[n]$ is the signal shape, perfectly known, its energy is also known and it is given by $E_s \triangleq \sum_{n=0}^{N-1} s^2[n] = N$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) given by $S_w(e^{j2\pi f}) = \sigma^2$.

- (1) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$; the pdf can be expressed either in scalar form and in vector form, please provide both.
- (2) Under the assumption that σ^2 is a-priori known, derive the Maximum Likelihood (ML) estimator of the amplitude A . Derive the pdf of the estimator $\hat{A}_{ML}$, derive the Cramér-Rao lower bound (CRB) and compare it with the Mean Square Error (MSE) of the ML estimator; establish whether the estimator is biased, consistent and efficient.
- (3) Under the assumption that σ^2 is also unknown, derive the joint ML estimator of the amplitude A and of the noise power σ^2 . Determine whether the two ML estimators are biased or not.



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Svolgimento

(1) Densità di probabilità (ddp) congiunta osservati.

$$X[n] \in N(A s[n], \sigma^2), \quad \{X[n]\}_{n=0}^{N-1} \text{ indipendenti} \Rightarrow \mathbf{X} \in N(\mathbf{A} \mathbf{s}, \sigma^2 \mathbf{I})$$

ddp in forma scalare:

$$f_x(\mathbf{x}) = \prod_{n=0}^{N-1} f_x(x[n]) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^N \exp \left(-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A \cdot s[n])^2 \right)$$

ddp in forma vettoriale:

$$f_x(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma^2 \mathbf{I}|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{A} \mathbf{s})^T (\sigma^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{A} \mathbf{s}) \right) = \frac{1}{(2\pi)^{N/2} (\sigma^2)^{N/2}} \exp \left(-\frac{1}{2\sigma^2} \|\mathbf{x} - \mathbf{A} \mathbf{s}\|^2 \right)$$

Logaritmo funzione di verosimiglianza:

$$\ln f_x(\mathbf{x}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A \cdot s[n])^2 = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{x} - \mathbf{A} \mathbf{s}\|^2$$

(2) Nell'ipotesi che σ^2 sia nota.

Stima ML di A:

$$\frac{d \ln f_x(\mathbf{x})}{dA} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} s[n] (x[n] - A \cdot s[n]) = 0 \Rightarrow \hat{A}_{ML} = \frac{\sum_{n=0}^{N-1} s[n] x[n]}{\sum_{n=0}^{N-1} s^2[n]} = \frac{1}{N} \sum_{n=0}^{N-1} s[n] x[n] = \frac{1}{N} \mathbf{s}^T \mathbf{x}$$

Essendo la stima una funzione lineare dei dati, che sono Gaussiani, anche la stima è Gaussiana, con valor medio e varianza calcolati nel seguito.

$$E\{\hat{A}_{ML}\} = \frac{1}{N} \sum_{n=0}^{N-1} s[n] E\{x[n]\} = \frac{1}{N} \sum_{n=0}^{N-1} s[n] A s[n] = A \cdot \frac{1}{N} \sum_{n=0}^{N-1} s^2[n] = A$$

quindi lo stimatore è non-polarizzato.

$$MSE\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = \text{var}\left\{ \frac{1}{N} \sum_{n=0}^{N-1} s[n] x[n] \right\} = \frac{1}{N^2} \sum_{n=0}^{N-1} s^2[n] \text{var}\{x[n]\} = \frac{\sigma^2}{N}$$

quindi lo stimatore è consistente.

$$\frac{d^2 \ln f_x(\mathbf{x})}{dA^2} = \frac{d}{dA} \left(\frac{1}{\sigma^2} \sum_{n=0}^{N-1} s[n] (x[n] - A \cdot s[n]) \right) = -\frac{1}{\sigma^2} \sum_{n=0}^{N-1} s^2[n] = -\frac{N}{\sigma^2}$$



$$CRB(A) = \frac{1}{-E \left\{ \frac{d^2 \ln f_x(\mathbf{x})}{dA^2} \right\}} = \frac{\sigma^2}{N} = MSE \{ \hat{A}_{ML} \}$$

quindi lo stimatore è efficiente.

(3) Nell'ipotesi che σ^2 sia incognita.

$\hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} s[n]x[n]$ è la stessa di prima perché non dipende dalla varianza del rumore, quindi ha le stesse proprietà di prima. La stima ML della varianza del rumore si ricava minimizzando rispetto a σ^2 la funzione di verosimiglianza “compressa”, che si ottiene inserendo nella funzione di verosimiglianza la stima ML di A al posto di A :

$$L_c(\mathbf{x}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - \hat{A}_{ML} \cdot s[n])^2$$

$$\frac{dL_c(\mathbf{x})}{d\sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{n=0}^{N-1} (x[n] - \hat{A}_{ML} \cdot s[n])^2 = 0 \Rightarrow \hat{\sigma}_{ML}^2 = \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - \hat{A}_{ML} \cdot s[n])^2$$

$$E \{ \hat{\sigma}_{ML}^2 \} = \frac{1}{N} \sum_{n=0}^{N-1} E \{ (x[n] - \hat{A}_{ML} \cdot s[n])^2 \}$$

$$E \{ (x[n] - \hat{A}_{ML} \cdot s[n])^2 \} = E \{ x^2[n] \} + E \{ (\hat{A}_{ML})^2 s^2[n] \} - 2s[n] E \{ x[n] \hat{A}_{ML} \}$$

$$E \{ x^2[n] \} = (E \{ x[n] \})^2 + \text{var} \{ x[n] \} = A^2 s^2[n] + \sigma^2$$

$$E \{ (\hat{A}_{ML})^2 \} = (E \{ \hat{A}_{ML} \})^2 + \text{var} \{ \hat{A}_{ML} \} = A^2 + \frac{\sigma^2}{N}$$

$$\begin{aligned} E \{ x[n] \hat{A}_{ML} \} &= E \{ x[n] \hat{A}_{ML} \} = E \left\{ x[n] \frac{1}{N} \sum_{i=0}^{N-1} s[i] x[i] \right\} = \frac{1}{N} \sum_{i=0}^{N-1} s[i] E \{ x[n] x[i] \} \\ &= \frac{1}{N} \sum_{i=0}^{N-1} s[i] (A^2 s[n] s[i] + \sigma^2 \delta[i-n]) = A^2 s[n] + \frac{\sigma^2}{N} s[n] = \left(A^2 + \frac{\sigma^2}{N} \right) s[n] \end{aligned}$$

$$\begin{aligned} E \{ \hat{\sigma}_{ML}^2 \} &= \frac{1}{N} \sum_{n=0}^{N-1} (A^2 s^2[n] + \sigma^2) + \frac{1}{N} \sum_{n=0}^{N-1} \left(A^2 + \frac{\sigma^2}{N} \right) s^2[n] - \frac{2}{N} \sum_{n=0}^{N-1} s[n] \cdot \left(A^2 + \frac{\sigma^2}{N} \right) s[n] \\ &= A^2 + \sigma^2 + A^2 + \frac{\sigma^2}{N} - 2 \left(A^2 + \frac{\sigma^2}{N} \right) = \sigma^2 - \frac{\sigma^2}{N} = \sigma^2 \left(1 - \frac{1}{N} \right) \neq \sigma^2 \end{aligned}$$

quindi lo stimatore della varianza del rumore è polarizzato ma asintoticamente non-polarizzato.